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Dear Drs. Feng, Frangou, and Murray,

We are excited to submit an Original Article entitled **Re-Examining Polygenic Risks and Psychosocial Environments: Are Gene-Environment Correlations Really That Pervasive?** for consideration to be published in *Psychological Medicine.* The authors, LiChen Dong, MS and James J. Li, PhD, are affiliated with Department of Psychology, Waisman Center, and Center for Demography on Health and Aging at University of Wisconsin-Madison, Madison, WI, USA.

Published in *Psychological Medicine*, a seminal meta-analysis by Kendler & Baker (2007) was one of the first to underscore genetic influences on measures of the environment. More recently, multiple studies, including one of our own (Dong et al., in press), have shown that polygenic risks for psychopathology may manifest indirectly through their influence on psychosocial environments. However, many of these studies have failed to consider the role confounders (e.g., genetic ancestry) may play in the observed polygenic-environment correlations (*r*GEs).

The current study aims to better understand *r*GEs by accounting for family-/population-level confounders. We generated polygenic scores based on the state-of-the-art externalizing GWAS (Karlsson Linnér et al., 2021) and tested their associations with adolescent social environments (i.e., parental relationships, delinquent peer affiliation, school connectedness) in the National Longitudinal Study of Adolescent to Adult Health. Utilizing a sample of twins and full siblings, we were able to estimate *r*GEs at both within- and between-family levels, so that we could partial out active/evocative genetic effects on the environments versus passive/confounding effects due to population stratification, assortative mating, and parental genotypes.

To our knowledge, our study is the first to incorporate a twin/sibling design and polygenic scores in studying genetic influences on the environment. Other strengths include our use of a large population-based sample of the United States, factor models to measure psychosocial environments, continuous shrinkage method to create portable polygenic scores, and Bayesian models to estimate attenuation of *r*GEs from between- to within-family levels of analysis.

To briefly summarize our findings, *r*GEs were observed with all environmental measures at the population level (i.e., among unrelated individuals), with between-family level analyses showing similar effects. However, at the within-family level, only rGE with parental relationships remained significant, while rGEs with other environments were greatly attenuated, likely due to confounding effects being controlled for. These results suggest that although rGEs are indeed quite ubiquitous, not all rGEs reflect causal effects of genes on the environment. Our findings also highlight the evocative effect of externalizing genetic risks on parent-child dynamics, which may contribute to an indirect pathway to externalizing behavior development. Finally, this study demonstrates the need for more rigorous, genetically informed designs that account for population- and family-level confounds in the investigation of gene-environment interplay.

This manuscript is formatted according to the *APA 7th Edition* and complies with all ethical standards in the treatment of participants. The manuscript is within the word limit and includes the recommended number of tables and figures. We followed open science practices for transparency. We also submitted Supplementary Tables to provide details regarding our measurement (i.e., factor) models for psychosocial environments. No material contained in this manuscript has been published or submitted for consideration for publication elsewhere.

LiChen Dong, M.S. is the first author, and James J. Li, Ph.D. is the senior author for this manuscript. Please direct correspondence to both authors, whose full address and contact information are listed below. Thank you for considering this manuscript for publication in *Psychological Medicine*. We look forward to hearing back about your decision.

Sincerely,

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